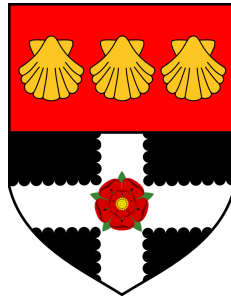


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Tipping points and early warning signals

with applications to geophysical data



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Changes made in response to examiners' reports

The examiners' reports, written in response to the original thesis submission, highlight many specific, necessary corrections to be made, many general suggestions for improvement and many comments on the overall structure and choice of subject matter. In order to deal with all of this feedback appropriately I have here extracted the individual suggestions from the examiners' report and the two preliminary reports:

Specific suggestions and corrections from the examiners' report

1. Very little is reported of the debate on tipping points in the Earth system. This needs to be expanded in order to put the results into a broader context.
2. The three kinds of tipping behaviours need to be described more thoroughly.
3. The study of scaling behaviour necessitates long datasets and large ensembles thereof. Such data are unavailable in this context and so the sensitivity of the results to the length of the signals and the size of the ensembles should be discussed.
4. The mathematical reasons behind the expected scaling behaviour should be more clearly expressed and discussed.
5. A definition of the periodogram is needed.
6. The thesis is unclear regarding the way noise is implemented in the dynamical systems of interest.
7. The author's understanding of the numerics of stochastic equations and Brownian motion is weak and must be reinforced.

8. The application presented in chapter ?? (tropical cyclone) does not correspond to the definition of a tipping point. That is, a tropical cyclone does not conform to a general bistable system.
9. The thesis is in need of a final chapter which will give a summary of the results, put them into a broader perspective and propose future research.

Suggestions and corrections from Report 1

The preliminary report of Examiner 1 catalogues 46 suggestions for specific changes, where these are sufficiently small that only a simple explanation, an additional reference or similar is required, these changes appear in red in this draft of the revised submission and are not discussed further. Amongst these 46 comments there are a few which necessitate further explanation:

- *Comment 6* refers to the dynamical system defined by a 2D system of ODEs, but which is not truly coupled and could be expressed as a single ODE.
- *Comment 12* refers to logarithmic binning.
- *Comments 13 and 16* refer to the use of two different methods for generating a red noise series.

Some of the suggestions in Report 1 are stated above in the nine points arising from the general examiners' report, such as that "it is not clear that indeed tropical cycles belong to the class of tipping point mechanism." Besides these, there are two other specific points:

1. Two of the early warning indicators used in the thesis are based on "scaling exponents", the PS and DFA scaling exponents. In addition, the scaling exponent of the auto-correlation function is also considered. Shortly, scaling is linked to power-law behavior in which there is no specific characteristic scale such that $f(x) = f(cx)/c^\alpha$ where α is the scaling exponent. Unfortunately, this seems to be overlooked, as the scaling exponent was estimated even when it is clear that there is no scaling, like when there is crossover (AR(1) process) and when the time series is "oscillatory" (like the Van-der-Pol oscillator). Indeed, it is possible to estimate an exponent in a range where there seems to be a power-law behavior but, from what I understood, the estimation of the scaling exponent was done without checking if indeed this is

the case. Even in the case of tropical cyclones where the author noted that there is a clear tidal 12 hours frequency and even proposed ways to filter out this frequency, eventually this frequency was not filtered out. In the last case, it is not clear that indeed there is scaling (probably not). The above may underlies the weak performance of the indicators. There are other technicalities related to the scaling which I list below.

2. The term "potential" is frequently used in the thesis. It seems to be used in inaccurate way. In physics "potential" is tightly linked to force through the relation $F = -\nabla(U)$ where the force is proportional to the second derivative of the location vector. In this context, a particle can be trapped in a potential well if its energy is smaller than the barrier's energy. However, in the examples considered in the dissertation, the potential is proportional to the first (time) derivative of the potential variable, e.g., $dx/dt = -dU/dx$. The "well" of U is not a real well as there is no relevant energy below which the "particle" is trapped. I would suggest the author to use the term "pseudo-potential" and to clarify that the potential he uses is not the same as the one used in physics in which a "well" is a well and a "barrier" is a barrier.

Suggestions and corrections from Report 2

Report 2 makes a number of specific suggestions:

1. Joshua does not provide a definition of the periodogram in his thesis.
A definition is now included. See definition ??, page ??.
2. There is no mention of how integration of stochastic differential equations are performed,
3. little discussion of the choice of parameters in many toy models (rate of change of parameters; intensity of noise),
4. nor clearly stated criteria for defining that the early warning indicator indeed flag the occurrence of a tipping point.
5. The previous part describes the early warning indicators in relation to specific [...] bifurcations of low-dimensional systems [...yet...] the application takes place on data which have nothing to do with bifurcation, but which he still classifies as “tipping behaviour”, with no further discussion. I believe that a logical step is missing there.

6. Finally, the thesis does not have conclusions section,
7. nor a sufficiently detailed discussion of tipping points in the Earth system.

List of papers and presentations

Papers

1. Prettyman, J., Kuna, T., and Livina, V. (2018). A novel scaling indicator of early warning signals helps anticipate tropical cyclones. *EPL (Europhysics Letters)*, 121(1):10002.

This paper is a result of the research detailed in chapter ??, particularly the development of the PS indicator. This paper also includes the results from section ?? where the one-dimensional indicators are applied to the sea-level pressure data.

2. Prettyman, J., Kuna, T., and Livina, V. (2019 - in print). Generalised early warning signals in multivariate and gridded data with an application to tropical cyclones. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, scheduled for publication July 2019.

This paper is a result of the research detailed in chapter ?? and sections ??-??, concerning early warning signals in multivariate and gridded data and the application of the methods to tropical cyclones. The model in section ?? is also included.

Presentations

- March 2016: Poster presentation given at *Mathematics of Planet Earth CDT second annual Jamboree*, University of Reading.
- October 2016: Seminar given at the National Physical Laboratory, Teddington.
- November 2016: Poster presentation given at conference *Time-series analysis in Environmental Science and Application to Climate Change*, University of Tromsø.

- March 2017: Poster presentation given at *Mathematics of Planet Earth CDT third annual Jamboree*, Imperial College London.
- April 2018: Poster presented at *EGU General Assembly 2018 (session - Mathematics of Planet Earth)*, Vienna.
- September 2018: Seminar given at conference *SIAM Mathematics of Planet Earth 2018*, Philadelphia.

Abstract

Tipping events in dynamical systems have been studied across many applications, often by measuring changes in variance or autocorrelation in a one-dimensional time series. In this thesis, existing techniques in tipping point analysis are reviewed and a novel method, the Power Spectrum indicator, is introduced. Methods for detecting early warning signals of tipping events in multi-dimensional systems are also reviewed and expanded and these techniques are applied to a variety of dynamical systems. An analytical justification of the use of dimension-reduction by empirical orthogonal functions, in the context of early warning signals, is provided and the one-dimensional techniques are also extended to spatially separated time series over a 2D field. The challenge of predicting an approaching tropical cyclone by a tipping-point analysis of the sea-level pressure time series is used as the primary example, and an analytical model of a moving cyclone is also developed in order to test predictions. We show that the one-dimensional power spectrum indicator may be used following dimension-reduction, or over a 2D field. We also show the validity of our moving cyclone model with respect to tipping-point indicators.

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List of abbreviations

- ACF — Autocorrelation function, see equation ??, page ??.
- ACF1 — Lag-1 autocorrelation function, see equation ??, page ??.
Used in the context of the ACF1 indicator, see definition ??, page ??.
- AR(p) — Autoregressive model of order p , see equation 1.11, page 6.
- DFA — Detrended fluctuation analysis, see definition ??, page ??.
Used in the context of the DFA indicator, see definition ??, page ??.
- EOF — Empirical Orthogonal Function, see section ??, page ??.
- EWS — Early warning signal, see section 1.3, page 19.
- PS — Power Spectrum, see equation ??, page ??.
Used in the context of the PS indicator, see definition ??, page ??.
- SLP — Sea-level pressure, see section ??, page ??.
- WS — Wind speed, see section ??, page ??.

Chapter 1

Introduction

This work is concerned with tipping points, also referred to as critical transitions, which are characterised by Kuehn [2011] as follows:

A non-mathematical working definition of a critical transition is an abrupt change in a dynamical system.

Scheffer [2009] uses the example of a canoe capsizing as the canoeist leans further over the edge. This example is one in which the *dynamical system* (the status of the roll of the canoe as it changes over time) has two stable states (upright and upside-down) and transitions *abruptly* from one to the other as a critical threshold is reached, in this case the point at which the centre of mass is sufficiently offset from the canoe's axis of vertical symmetry that it becomes unstable.

In order to take this thesis beyond a superficial study of the canoeist example, we introduce in sections 1.1 and 1.2 the concepts of dynamical systems, time series analysis and tipping points, including some common examples. In section 1.3 we introduce the foundational ideas of early warning signals, which may be used to predict tipping points. In section 1.4 we outline the structure and key points of the remainder of this thesis.

1.1 Dynamical systems and time series analysis

The term *dynamical system* is used in pure mathematical analysis, medicine, climatology, and other disciplines. For the purposes of the physical sciences, a dynamical system is defined as "a particle or ensemble of particles whose state varies over time" [Nature, 2019]: that is, almost any measurable quantity or quality of almost anything that changes over

time. The mathematical field of Dynamical Systems more specifically defines a dynamical system as a triple $(\mathcal{S}, \mathcal{T}, \Phi)$, where $\mathcal{S}$ is the state space, $\mathcal{T}$ is the parameter space and the map

$$\Phi : (\mathcal{S} \times \mathcal{T}) \rightarrow \mathcal{S} \quad (1.1)$$

is the evolution determining the dependence of the state $s \in \mathcal{S}$ on the parameter $t \in \mathcal{T}$ [Brown, 2018]. The state of the system as the parameter changes is called the *trajectory* of the system and $(\mathcal{S} \times \mathcal{T})$ is known as the trajectory space.

The majority of the influence on this thesis is tipping point research relating to real physical systems changing over time, such as the change in atmospheric oxygen concentration over a period of years [Livina et al., 2015]. In chapter ?? the tipping point analysis techniques developed and studied throughout the previous chapters are applied to observed sea-level pressure measurements as they change over a period of days. Therefore, in the context of this thesis, $t \in \mathcal{T}$ is typically the "time" parameter and $\mathcal{T}$ is typically the set of real numbers $\mathbb{R}$, or a discrete subset (e.g. $\{0, 0.1, 0.2, \dots\}$). Hence, the trajectory is the state of the system as it changes with time. This brings the mathematical definition of a dynamical system closer to the physical sciences definition. In this thesis we generally distinguish an analytically defined *dynamical system*, where the map Φ is often defined by a set of differential equations [Robinson, 2012], from a *physical system*, such as the roll angle of a canoe or the sea-level pressure at a specific geographical location. However, the two concepts overlap where a physical system is presumed to be governed by (possibly unknown) analytical equations, or when an analytical dynamical system is used to model the behaviour of a physical variable.

A simple example of a dynamical system is based on the quadratic map

$$f(x) = 4x(1 - x). \quad (1.2)$$

Here, the state is $x \in \mathbb{R}$, the 'time' parameter is $n \in \mathbb{N}$ and the dynamics are governed by the map

$$\Phi : (x, n) \rightarrow f^n(x), \quad (1.3)$$

denoting the function f iterated n times. The trajectory of this dynamical system, with the initial condition $x_0 = 0.4$, is shown in figure 1.1. The series of values $X = \{x_0, x_1, \dots\}$ is, if we continue to refer to the parameter n as 'time', a *time series*, a collection of observations made over time [Chatfield, 2016]. These observation can be the hourly price

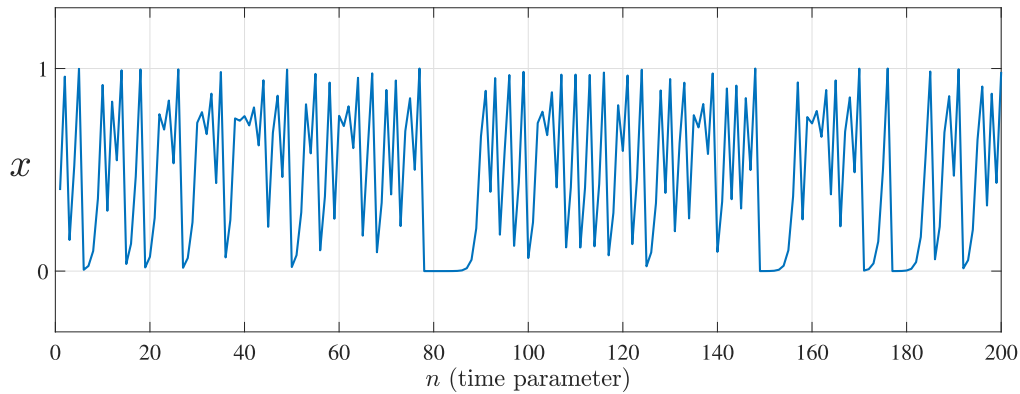


Fig. 1.1 The quadratic map (equation 1.2) iterated 200 times with initial value $x_0 = 0.4$.

of a commodity, the yearly global average temperature deviation or, in this case, the state x of the dynamical system in equation 1.3.

In chapters ?? and ?? several dynamical systems defined by analytical equations are studied in order to detect or predict tipping points which occur in those systems. An instructive example, which is referred to throughout this thesis, is the *double-well potential* system, described by the differential equation

$$\dot{z} := \frac{dz}{dt} = -\frac{\partial}{\partial z} (z^4 - 2z^2), \quad (1.4)$$

where $z(t)$ is the system state variable and t is the time parameter. The right-hand side could be written more concisely as $4z - 4z^3$ but expressing it as a derivative allows us to see the shape of the *generalised potential function* $U(z) = z^4 - 2z^2$, which is illustrated in figure 1.2a. The generalised potential function, in this case, describes two *wells* into which the system state falls. The term “potential function” in physics has a specific meaning relating to the energy potential of a particle but in dynamical systems it is useful to refer to the “generalised potential function”, as we have done here, in the sense that its minima correspond to the most probable states of the system [Nicolis and Nicolis, 2007]. In this sense the reciprocal of the generalised potential function is related to the probability distribution function of system states, but it is more convenient for our purposes to refer only to the generalised potential function.¹

¹All instances of the term “potential function” from this point onwards in the manuscript have now been changed to “generalised potential function”, consistent with the phrasing in Nicolis and Nicolis [2007], etc..

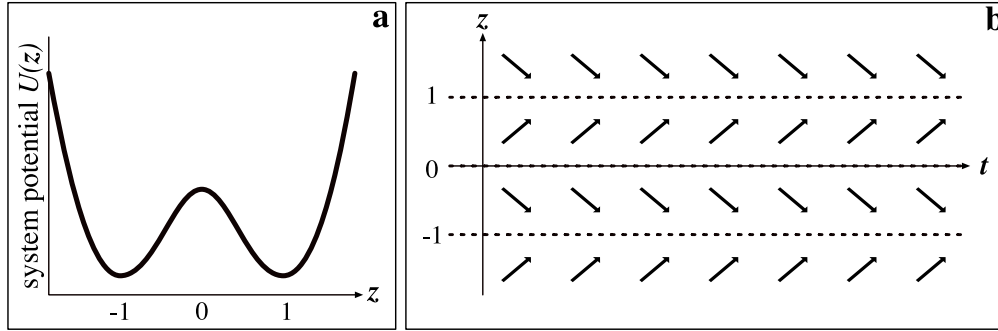


Fig. 1.2 (a) The shape of the double-well generalised potential function $U(z) = z^4 - 2z^2$, and (b) the trajectory of the dynamical system described by equation 1.4. The dashed lines represent the equilibria $z = -1, 0, 1$.

As t changes, so too does the system state z unless $\dot{z} = 0$, which occurs at the *equilibria*, the states $z = -1, 0, 1$, which are also the solutions to

$$\frac{\partial}{\partial z} (z^4 - 2z^2) = 0. \quad (1.5)$$

Thus, we are able to understand the dynamical system in terms of its equilibria. At $z = 0$ there is an *unstable* equilibrium where $U'' = 12z^2 - 4 = -4 < 0$ and so if z is close to, but not equal to zero, the evolution of the system (the change as t increases) will move away from this state, along the arrows of the trajectory in figure 1.2b, towards the *stable* equilibria $z = \pm 1$, where $U'' = 12z^2 - 4 = 8 > 0$. Because of the shape of the generalised potential function, these stable equilibria are known as wells or basins of attraction. Any state $z \neq 0$ will evolve towards either one of these wells and will become asymptotically close to $z = \pm 1$.

1.1.1 Stochastic dynamical systems

When attempting to model a physical system, a purely deterministic equation may not be appropriate [Shumway and Stoffer, 2017] and so it is useful to couple a dynamical system with a stochastic process to model uncertainty or unknown short-scale processes. The stochasticity can also induce tipping in the dynamical system — we note that in the double-well potential system introduced above, the system variable z heads towards one of the potential wells (one of the stable states of the system) and remains there: this is not a

suitable model for a physical system such as the canoe example, which tips from one state to another.

The most basic example of a discrete-time stochastic process, for our purposes, is independent Gaussian white noise, that is, a time series where each element is sampled independently from a zero-mean Gaussian distribution. In several systems throughout this thesis we introduce additive noise [Honerkamp, 1994] so that a system

$$\frac{dx}{dt} = f(x), \quad (1.6)$$

becomes

$$\frac{dx}{dt} = f(x) + \eta_t, \quad (1.7)$$

where η is a Gaussian white noise process. Adding a white noise process to the double-well potential system in equation 1.4, we get a stochastic dynamical system described by the stochastic differential equation

$$\dot{z} = -\frac{\partial}{\partial z} (z^4 - 2z^2) + \sigma \eta_t, \quad (1.8)$$

where the constant σ alters the variance of the noise term. In this system the overall evolution is still towards the two wells of attraction, and we note that a particle starting exactly in the state $z = 0$ will no longer remain on the unstable equilibrium as perturbations from the noise term will move the particle slightly away from zero, after which point it will be attracted into one of the two wells. In figure 1.3 the trajectory of the system is shown with initial condition $z = 0$ for small noise ($\sigma = 0.4$) and for large noise ($\sigma = 0.7$). Note that when the noise is sufficiently large a particle is able to cross over the $z = 0$ point, the boarder between the two wells of attraction, and move into the other well. This abrupt, noise-induced transition from one stable state to another is one of the several types of tipping, which will be described in section 1.2. Significantly, this transitional tipping point is not caused by a change in the shape of the generalised potential, as is the case for other examples of tipping points.

Besides Gaussian white noise, we may introduce many different types of stochastic processes: Brownian motion, fractional ARIMA processes, the Ornstein-Uhlenbeck process, etc. [Gardiner, 1994; Box et al., 2008; Chatfield, 2016; Shumway and Stoffer, 2017]. We make a note of the *discrete random walk*, which is simply the cumulative sum of a

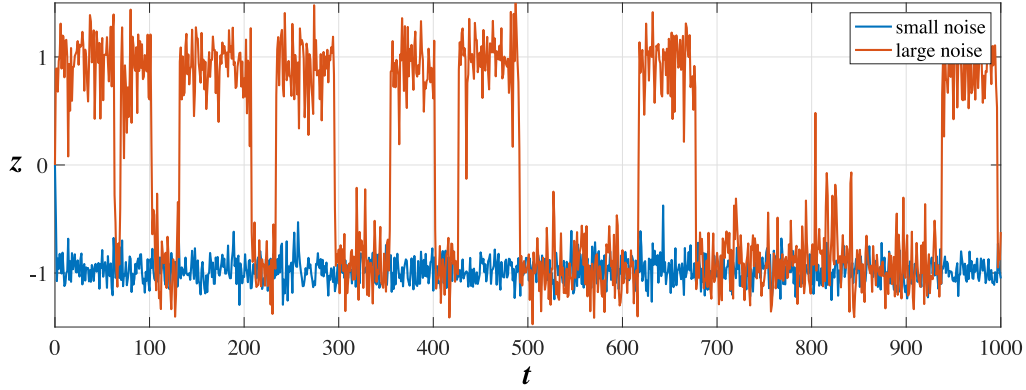


Fig. 1.3 Time series of the possible trajectory of the stochastic double-well potential system described by equation 1.8 with small noise (blue line, $\sigma = 0.4$) and large noise (red line, $\sigma = 0.7$).

white noise series, thus:

$$x_t = \sum_{j=1}^t \eta_j, \quad (1.9)$$

where η is Gaussian white noise. This could also be defined recursively by the equation

$$x_t = x_{t-1} + \eta_t, \quad (1.10)$$

with initial condition x_0 . A more general stochastic process with a similar form is the order- p autoregressive process, or $AR(p)$ process, described by the equation

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + \eta_t, \quad (1.11)$$

where $\phi_1, \phi_2, \dots$ are constant parameters. Figure 1.4 shows the time series of a random walk, and an $AR(2)$ process with $\phi_1 = 0.8$, $\phi_2 = -0.2$. These two time series appear different, we note that the $AR(2)$ process appears to be stationary (the mean does not change over time) whereas the random walk does not.

1.1.2 Methods of analysis

There exist many methods for the analysis of the time series which arise both from measurements of physical systems and from analytically defined dynamical systems [Box et al., 2008; Chatfield, 2016]. Besides the mean and sample variance, we define the sample

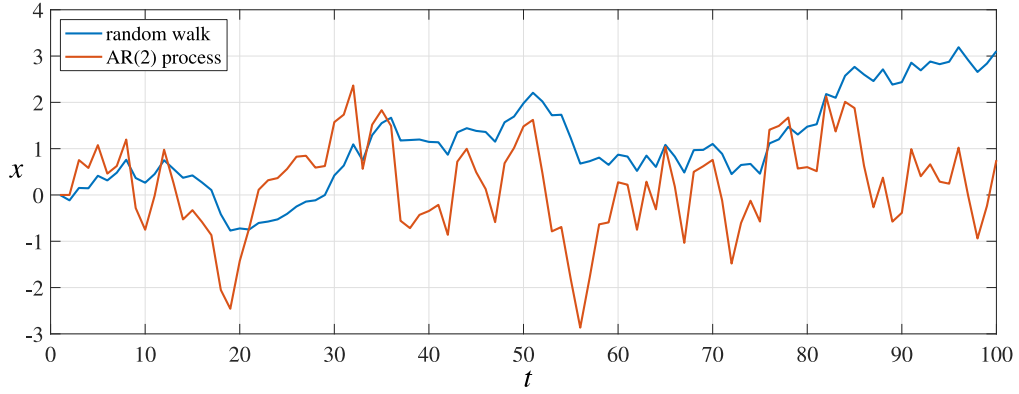


Fig. 1.4 Time series of a random walk (blue line), and an AR(2) process with $\phi_1 = 0.8$, $\phi_2 = -0.2$ (red line).

autocorrelation function (ACF), a measure of the correlation between a time series and a delayed copy of itself. Since, in tipping point research, we are concerned with abrupt changes in systems, autocorrelation is frequently used to analyse time series of systems in which tipping has occurred or may occur [Held and Kleinen, 2004; Scheffer et al., 2009]. In chapter ?? the *ACF indicator* is introduced as a means to quantify how lag-1 autocorrelation changes over time.

Definition 1.1.1 (Autocorrelation function) *The ACF with lag k of a time series $X(n)$ is given by*

$$ACF_k(X) = \frac{1}{(N-l)s^2} \sum_{j=1}^{N-l} (X_j - \bar{X})(X_{j+k} - \bar{X}), \quad (1.12)$$

where $\bar{X}$ and s are the sample mean and sample standard deviation of X .

The lag-1 ACF of the random walk and the AR(2) process in figure 1.4 are found numerically to be 0.9995 and 0.6658 respectively, giving just one quantitative measure of the difference in behaviour between the two processes. The ACF of a Gaussian white noise **has expected value zero**, since each point is sampled independently and so there is no correlation between them.

Many methods of time series analysis involve transforming the series from the *time domain*, in which the system state is a function of time, into the *frequency domain*, in which the amount of the series lying within a frequency band is a function of frequency [Chatfield, 2016], thereby obtaining the power spectrum for the particular time series, which we define here:

Definition 1.1.2 (Power Spectrum) *The Power Spectrum of a time series $x(t)$ is the Power Spectral Density (PSD) of $x(t)$ considered as a function of frequency f . We denote the power spectrum by $S_x(f)$. Alternative formulations consider the PSD as a function of angular frequency $\omega = 2\pi/f$.*

For a time series $x(t)$ with Fourier transform

$$\hat{x}(f) = \int_{-\infty}^{\infty} x(t)e^{-2\pi ift} dt, \quad (1.13)$$

the power spectrum $S_x(f)$ is given by the modulus squared of the Fourier transform,

$$S_x(f) = |\hat{x}(f)|^2. \quad (1.14)$$

Now we have defined the power spectrum we are in a position to define *white noise*, which was introduced in equation 1.7 (section 1.1.1) and is used throughout this thesis.

Definition 1.1.3 (White noise) *A white noise process w_t is a random signal with equal power spectral density at all frequencies, that is, the power spectrum of the process is a constant:*

$$S_w(f) = |\hat{w}(f)|^2 = \left| \int_{-\infty}^{\infty} w_t e^{-2\pi ift} dt \right|^2 = C \quad (1.15)$$

for positive constant C [Robinson, 2012].

A corollary of the definitions of the white noise series is that

$$\begin{aligned} |\hat{w}(f)|^2 &= C \\ |\hat{w}(f)| &= \sqrt{C} \\ \hat{w}(f) &= \sqrt{C}e^{i\theta f}, \end{aligned} \quad (1.16)$$

for some $\theta \in [-\pi, \pi]$. In this thesis only Gaussian white noise is considered, in which case the distribution of the values of w_t will be Gaussian, although many other distributions of values could also result in a constant power spectrum. In computational examples and experiments throughout this thesis, whenever a white noise series is required, a discrete-time series of length N is generated simply by using the Matlab random number generator function `randn(1,N)`, which returns N independent identically distributed values sampled from a normal distribution with mean zero and $\sigma^2 = 1$.

An alternative definition of the power spectrum, or PSD, is based on the auto-correlation of the time series $x(t)$. The Wiener-Khinchin theorem states that the power spectrum $S_x(f)$ is equal to the Fourier transform of the auto-correlation function [Chatfield, 2016]. In this way we see that the power spectrum and the auto-correlation function are closely related, which, in chapter ??, will motivate the introduction of power spectrum scaling exponent as a tool for tipping point detection [Prettyman et al., 2018].

Theorem 1.1.1 (The Wiener-Khinchin Theorem) *For a time series $x(t)$ the auto-correlation function $R_x(\tau)$ given by*

$$R_x(\tau) = \mathbf{E}[x(t)x(t + \tau)] \quad (1.17)$$

may be expressed in terms of the power spectral density thus

$$R_x(\tau) = \int_{-\infty}^{\infty} S_x(f) e^{2\pi i \tau f} df, \quad (1.18)$$

which is the inverse Fourier transform [Chatfield, 2016].

If the terms R and S in equation 1.18 satisfy the conditions of Fourier inversion, i.e. that they are absolutely integrable, then we may state the theorem as

$$S_x(f) = \int_{-\infty}^{\infty} R_x(\tau) e^{-2\pi i \tau f} d\tau = \hat{R}_x(f). \quad (1.19)$$

This Fourier transform formulation is often used as the definition of the power spectrum [Ricker, 2012], and is similarly expressed in the case of a discrete-time process $x(n)$ as follows:

$$S_x(f) = \sum_{k=-\infty}^{\infty} R_x(k) e^{-2\pi i f k}, \quad (1.20)$$

where the discrete time autocorrelation is given by

$$R_x(k) = \mathbf{E}[x(n)x(n - k)]. \quad (1.21)$$

In chapters ??, ?? and ?? of this thesis, where autocorrelation is used as a tool in tipping point detection and is applied to finite, discrete time series, we use the normalised autocorrelation function

$$ACF_k(x) = \frac{1}{\sigma^2} R_x(k) = \frac{\mathbf{E}[x(n)x(n - k)]}{\mathbf{E}[x(n)^2] - \mathbf{E}[x(n)]^2} \quad (1.22)$$

where σ^2 is the variance of $x(n)$. Moreover, the sample mean (hence sample variance) must be used as an approximation to the expected value so that we end with the sample autocorrelation function $ACF_k(x)$ as defined in definition 1.1.1, equation 1.12. Similarly, the power spectrum, which is calculated in equation 1.20 as an infinite sum of autocorrelations, must be calculated otherwise if the methods dependent on spectral analysis are to be applied to finite time series. An straight-forward estimation of the power spectrum, using the discrete-time fast Fourier transform, is known as a *periodogram* [Welch, 1967; Oppenheim, 1999].

Definition 1.1.4 (Periodogram) *The periodogram $\hat{P}(f')$ of a time series x_n of length N , as used throughout this thesis, is defined by the equation*

$$\hat{P}(f') = \frac{1}{N} \left| \sum_{n=0}^{N-1} x_n e^{-2\pi i f' n} \right|^2, \quad f' = 0, \frac{1}{N}, \frac{2}{N}, \dots, 1, \quad (1.23)$$

where f' is a discrete variable equivalent to the frequency f in the power spectrum (definition 1.1.2). The summation term is the fast Fourier transform as implemented in Matlab [Frigo and Johnson, 1998, 2005]. In this thesis, for the purposes of using the periodogram as a tool for tipping point detection, we consider only the one-sided periodogram, that is, the periodogram $\hat{P}(f')$ for the frequency range

$$f' = 0, \frac{1}{N}, \frac{2}{N}, \dots, \frac{1}{N} \left\lfloor \frac{N}{2} \right\rfloor, \quad (1.24)$$

since the periodogram, as defined here, is symmetric about $f' = 1/2$ [Fulop and Fitz, 2006].

Other methods exist to estimate the power spectrum, such as Welch's method [Welch, 1967] and the multitaper method [Thomson, 1982], both of which are based on the periodogram and mentioned in chapter ??.

Figure 1.5 shows the periodogram of a Gaussian white noise series and a random walk (both of length 10^6). The Gaussian white noise series η_n is simply a series of independent values sampled from a Gaussian distribution, its autocorrelation function, from equation 1.21, is ths given by

$$R_\eta(k) = \mathbf{E}[\eta(n)\eta(n-k)] = \begin{cases} 0, & k \neq 0 \\ 1, & k = 0 \end{cases}, \quad (1.25)$$

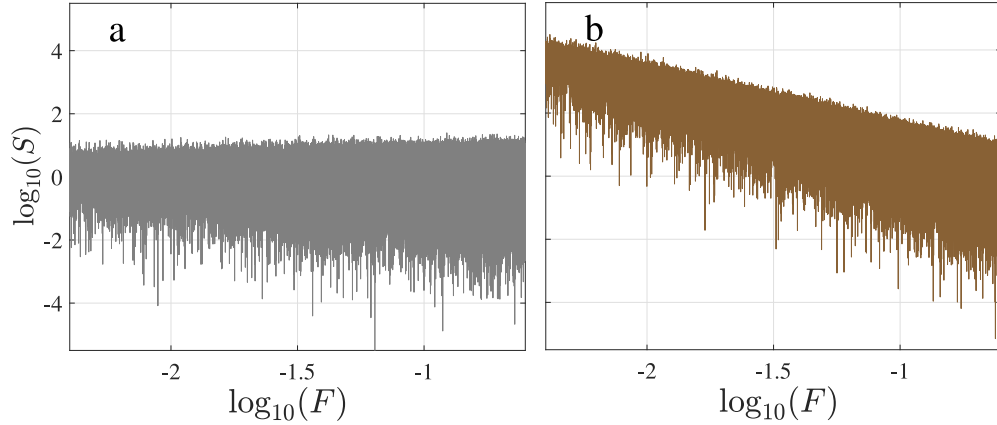


Fig. 1.5 The periodogram of a Gaussian white noise series (panel **a**) and a random walk (panel **b**) both of length 10^6 . The logarithm of the spectral power S is plotted over the logarithm of the frequency F in both cases in the frequency range $10^{-2} < F < 10^{-1}$.

and the power spectrum as given by the Wiener-Khinchin theorem (equation 1.20) becomes

$$S_{\eta}(f) = \sum_{k=-\infty}^{\infty} R_{\eta}(k) e^{-2\pi i f k} = R_{\eta}(0) e^0 = 1. \quad (1.26)$$

The power spectrum of the Gaussian noise is thus constant and is therefore white noise according to definition 1.1.3. The red noise (random walk) series in figure 1.5 is simply the cumulative sum of a Gaussian white noise series. In continuous time, red noise (or a Wiener process) is defined as a process W_t such that the derivative is a Gaussian process:

$$\frac{d}{dt} W_t = \eta_t. \quad (1.27)$$

Or, conversely, as the integral of a Gaussian process. Using this definition we may compute the power spectrum of a Wiener process using the Fourier transform identity that if $\hat{x}(f)$ is the Fourier transform of $x(t)$, then

$$\widehat{\frac{d}{dt} x(t)} = (2\pi i f) \hat{x}(f). \quad (1.28)$$

For the Wiener process this identity becomes

$$\widehat{\frac{d}{dt} W_t} = (2\pi i f) \hat{W}_t. \quad (1.29)$$

or,

$$(2\pi if)\hat{W}_t = \hat{\eta}_t = e^{i\theta f} \quad (1.30)$$

for some $f \in [-\pi, \pi]$, as shown in equation 1.16. Therefore the power spectrum is given by

$$S_W(f) = |\hat{W}_t|^2 = \left| \frac{e^{i\theta f}}{2\pi if} \right|^2 = \frac{1}{4\pi^2} f^{-2}. \quad (1.31)$$

This negative scaling factor of -2 can be observed in the log-log plot in figure 1.5b. In chapter ?? we introduce the power spectrum scaling indicator [Prettyman et al., 2018] which estimates this scaling factor by measuring the gradient of the log-log plot of the periodogram. Here we remark that due to the Wiener-Khinchin theorem (theorem 1.1.1) there exists a close relationship between the power spectrum and the autocorrelation. Therefore, by analysing the periodogram, we are able to introduce a method for detecting tipping points given that the autocorrelation is already widely used for this purpose. Also in chapter ??, we describe yet another time series analysis technique: detrended fluctuation analysis [Kantelhardt et al., 2001], and we provide a comparison of the different techniques for early warning signals, which are based on scaling properties of time series.

1.1.3 Higher dimensions

The time series and dynamical systems described above have only a single-valued system variable x , which is required by techniques such as the ACF or the power spectrum. In chapter ?? we expand the scope of the thesis to include higher-dimensional systems such as

$$\begin{aligned} \dot{r} &= \mu r - r^3, \\ \dot{\theta} &= 1 + r^2, \end{aligned} \quad (1.32)$$

which has two variables (in this case, r and θ). In a physical system, **this higher-dimensionality** could correspond to measuring several different physical variables, or measuring the same variable at several discrete locations in space. It may be possible only to consider a single critical variable such as temperature. Otherwise it is also possible to combine several time series into one, for example using Empirical Orthogonal Functions (EOFs), which we discuss in section ??, or to use a technique similar to ACF but expanded to higher dimensions [Williamson and Lenton, 2015], which we discuss in section ?. As a further alternative we introduce, in section ??, the possibility of analysing several time

series simultaneously using 2D plots. All of these ideas are then applied to a practical geophysical problem in chapter ??.

1.2 Bifurcations and tipping points in dynamical systems

In this section we introduce the concept of a bifurcation, which will be closely related to the concept of a tipping point. Introduced by Poincaré and meaning a split or fork, the general term bifurcation does not have a specific meaning, although specific types of bifurcation are well defined in linear dynamical systems. It is understood to mean a change in the parameters of a dynamical system which causes a significant qualitative change in the system's behaviour [Devaney, 2003; Blanchard et al., 2006]. The dynamical systems and stochastic processes seen in the previous chapter do not contain bifurcations in any established sense, although the change in the parameter (time) does cause changes in the systems. In the double-well potential system in figure 1.3 (large noise) switches from one well to another but this cannot be called a bifurcation since the change occurs due to perturbations from the noise term, not due to a change in the parameter t (that is, it could potentially happen at *any* time t). We can introduce a bifurcation into the system by varying the coefficient of z^2 in equation 1.8 so that the shape of the generalised potential function U changes:

$$U = z^4 + \left(3 - \frac{t}{200}\right) z^2. \quad (1.33)$$

When $t = 1000$ we have the same double-well potential system as in equation 1.8 with $U = z^4 - 2z^2$. However, when $t < 600$ the shape of the generalised potential function is qualitatively different: only a single potential well exists. Thus we have introduced a parametric forcing into the system. Such parametrically forced systems are common in the study of bifurcations [Strogatz, 2014] since it is useful to consider the nature of the equilibria of a system at various values of the forcing parameter, in particular those values when the bifurcation occurs. In this chapter and the next we study parametrically forced systems which exhibit *genuine bifurcations*, where by varying the parameter the shape of the generalised potential function is changed and the equilibria of the system either appear, disappear or change in nature. This, however, is not the only way in which a tipping point may occur in an analytically defined dynamical system: we follow authors such as Scheffer et al. [2009] in defining a tipping point as any abrupt qualitative change in

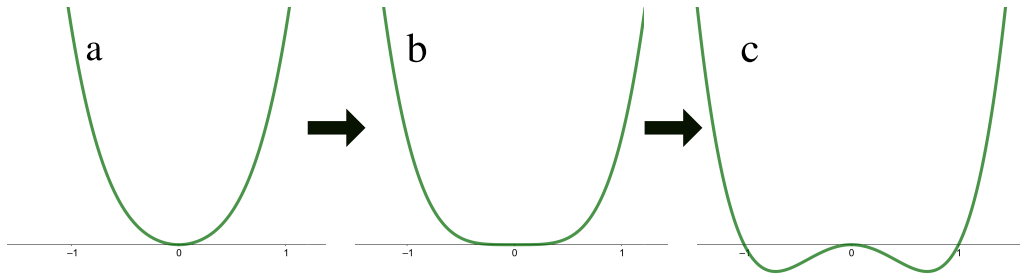


Fig. 1.6 Schematic of the changing shape of the generalised potential function U in equation 1.33. U is shown for $t < 600$ where the shape is a "single-well" (panel **a**); for $t = 600$ where the bifurcation occurs (panel **b**); and for $t > 600$ when the "double-well" shape has developed (panel **c**).

the dynamical system, which may include a shift in the system state caused purely by the inherent stochasticity of a system, referred to as *noise-induced tipping* Livina et al. [2011].

The evolution of the generalised potential function U in equation 1.33 is shown in figure 1.6. The time series in figure 1.7 show two possible trajectories of the system. The bifurcation at $t = 600$ is another example of a tipping point, distinct from the noise-induced transition seen in figure 1.3. In chapter ?? this same dynamical system is used to create multiple time series on which we test the tipping point techniques described in that chapter and successfully identify the tipping event at $t = 600$. We note that the bifurcation in figure 1.7 appears to happen gradually in reference to the $0 \leq t \leq 1000$ scale presented in the plot: in each of the trajectories we are able to see the increasing variance at $t = 400$, and it is not clear that a double-well potential has developed until $t > 700$. The reference to an "abrupt change" in the definition of a tipping point [Kuehn, 2011] is in comparison to large, long-scale deterministic components in the system, which are not present in this simple system for $t < 600$. Despite this, systems such as this are commonly used as examples in tipping point analysis [Scheffer et al., 2009; Kuehn, 2011] because the principle concepts would remain the same if a long-scale deterministic component were introduced. For example, the stable equilibria might be described by a sine wave with period 2000, rather than the line $z = 0$; plotted on the time scale $-10^5 < t < 10^5$ the tipping at $t = 600$ will be apparently "abrupt".

1.2.1 Defining a "tipping point"

The casual definition of a tipping point offered by Kuehn [2011], "an abrupt change in a dynamical system", suggests that tipping points are not restricted to a particular type of

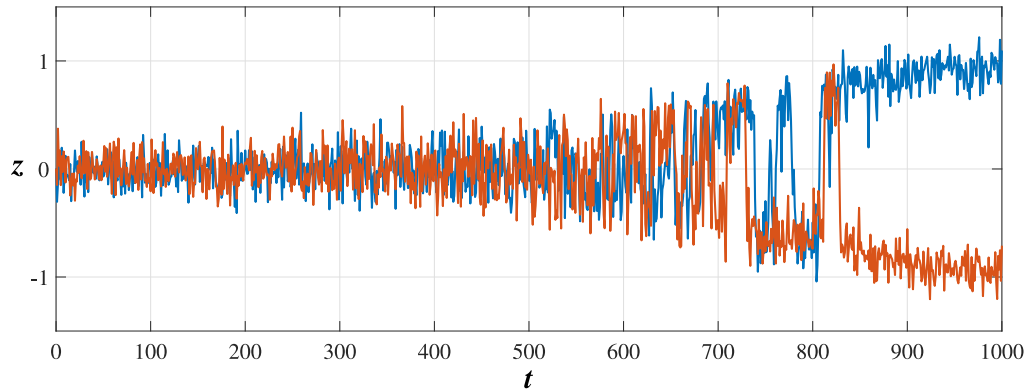


Fig. 1.7 Time series of two possible trajectories of the stochastic double-well potential system with changing generalised potential function described by equation 1.33. The equation is the same in both cases, with noise level $\sigma = 0.4$.

change in a dynamical system, either a bifurcation or a transition from one state to another, but any system in which a sudden change can be observed.

Because the definition is very broad, tipping points are studied in a variety of different contexts using a variety of different methods [Scheffer, 2009]. Some areas of research in which tipping points are studied include finance [Scheffer et al., 2009], ecology [Scheffer et al., 2001], biochemistry [Bagowski and Ferrell Jr, 2001] and climate [Lenton et al., 2008]. Despite the breadth of the subject, many systems exhibiting tipping points are mathematically related, that is, when these systems are close to the tipping point they share certain general properties in their dynamics which arise regardless of their individual natures [Scheffer et al., 2009]. These shared properties, enumerated by Scheffer [2009] are summarised by Kuehn [2011] in the following list:

1. An abrupt qualitative change in the dynamical system occurs.
2. The change occurs rapidly in comparison to the regular dynamics.
3. The system crosses a special threshold near a transition.
4. The new state of the system is far away from its previous state.
5. There is small noise in the system i.e. the data has a major deterministic component with small “random fluctuations”.
6. The system recovers slowly from perturbations (“slowing down”).

7. The variance of the system increases as the transition is approached.
8. The noisy fluctuations become more asymmetric.
9. The autocorrelation increases before a transition.

The point of interest is the third point: “The system crosses a special threshold near a transition.” since the nature of the “special threshold” determines the nature of the tipping point and therefore which techniques are used to analyse it.

The idea of the critical threshold comes from catastrophe theory which has in the past been criticised for an unscientific application of the cusp catastrophe to a huge variety of social and biological systems [Zahler and Sussmann, 1977], including the fight reflex of dogs and the psychological states of *anorexia nervosa* sufferers [Zeeman, 1977], all of which represent tipping points in a broad sense. Despite this, the cusp catastrophe (or fold bifurcation) is widely used as an example in tipping point research [Scheffer et al., 2001; van Nes and Scheffer, 2005; Van Nes and Scheffer, 2007; Kuehn, 2011]. Scheffer et al. [2009] note that the cusp catastrophe model is “now considered to capture the essence of shifts at tipping points in a wide range of natural systems” and also that critical thresholds for tipping points correspond to bifurcations. In figure 1.8 three tipping points are illustrated: a non-catastrophic transition in which the system state changes smoothly; a cusp catastrophe in which the system undergoes a bifurcation; and a transition from one stable state to the another. In the first two cases (panels **a** and **b**) the same forcing is used and the same final system state is reached. Given a sufficiently large forcing, therefore, it would not be possible to tell the difference between the two scenarios. We use this idea in chapter ?? to justify the use of the tipping point analysis in a novel system: the approach of a tropical cyclone. This system is possibly not an example of a two-state system with a genuine bifurcation or transition, but it is possible that the same analysis will work if the same sort of forcing is present as if it *were* a bifurcation. Indeed, Ashwin et al. [2012] note, in the context of tipping point analysis, that tipping points do not only include bifurcations.

A comparison of the transitions in figure 1.8 panels **b** and **c** demonstrates the two possible ways to shift between alternate stable states [Scheffer et al., 2001]: either a slight change in the system conditions, possibly a parameter value, close to the bifurcation point; or a sufficiently large perturbation due to noise which is able to push the system over the boarder to the other attractor well. Ashwin et al. [2012] describe three types of tipping with applications to climate: bifurcations, noise-induced tipping, and rate-induced tipping.

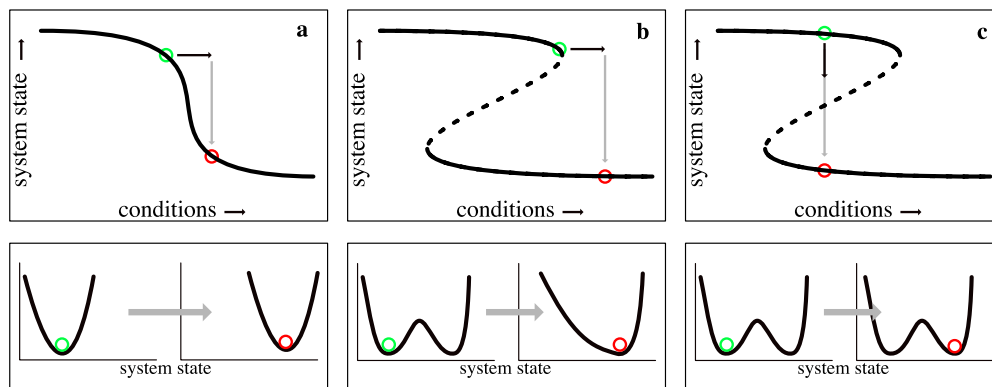


Fig. 1.8 Three distinct tipping points: a non-catastrophic transition (a), a cusp catastrophe bifurcation (b) and a transition between stable states (c). In each case the small black arrow represents a small forcing and the grey arrow represents the resulting change in the system state. The attractor is shown in the top panel and the system generalised potential function is shown beneath.

Livina et al. [2011] identify three types of system changes which may be the cause of a tipping event:

1. A forced transition, where the generalised potential function stays the same shape, but is moved, so that there is a trend (or possibly a sudden jump) in the time series.
2. A genuine bifurcation, where the shape of the generalised potential function is changed.
3. A noise-induced transition, where the system shifts from one stable state to another due to a large initial perturbation.

We note that these three scenarios are represented in figure 1.8 (a, b and c resp.) where the forcing of the system state in figure 1.8c is noise-driven. Rate-induced tipping, which is not represented in figure 1.8, occurs as a result of the rate of change of a system parameter (or system conditions).

Our working definition of a tipping point refers to an abrupt change in a dynamical system, which may also refer to a change in the system potential or underlying mechanics, besides simply the measurable state of the system as the three examples in figure 1.8 suggest. Livina et al. [2011] study a system such as

$$\dot{z}(t) = -\frac{\partial}{\partial z} (z^4 - 2z^2) + \sigma(t)\eta, \quad (1.34)$$

where η is white noise and $\sigma(t)$ is a parameter which varies with time and governs the size of the noise. In this example, where $\sigma(t)$ is made to decrease from 2 to close to zero, a system with a double-well generalised potential function effectively becomes a single-well system as the noise term becomes so small that it is practically impossible for the system to change state, although the actual shape of the generalised potential function does not change. In this case, the qualitative change in the system is not a change in the system state, but a decrease in the probability that the system state will change. If this decrease in probability is considered “abrupt” in comparison to the long-term dynamics, this system could be said to exhibit a tipping point at around the time that $\sigma(t)$ starts decreasing, even though the system state may not apparently change at that point. Indeed, this type of tipping point may only become apparent from the nature of the equations that govern the system. If presented with time series data alone, it may be difficult, or impossible, to judge whether a tipping has occurred.

1.2.2 Varieties of bifurcations

Although we have now defined the term *tipping point* very generally as any abrupt change in a dynamical system [Kuehn, 2011], and the early-warning techniques introduced in chapter ?? are applied to systems exhibiting such generally-defined tipping, we will focus, in chapter ??, on examples of parametrically forced bifurcations in order to facilitate analysis.

The term *bifurcation*, like *tipping point*, is used very generally and without a specific definition [Scheffer, 2009]. The word is literally a description of the pitchfork bifurcation depicted in figures 1.6 and 1.7: the single stable equilibrium splits in two (bifurcates) at the critical point. Rather, bifurcations are understood in terms of specific, named examples of bifurcations, all of which generally involve the creation or decay of stable equilibria or, in two or more dimensions, stable limit cycles, or, in three or more dimensions, chaotic attractors. In this section we give brief descriptions of the bifurcations which are used throughout this thesis.

The pitchfork bifurcation

Super-critical (already studied) and Sub-critical

The Hopf bifurcation

The homoclinic bifurcation

1.3 Early warning signals of tipping points

Closely related to the idea of tipping points is the idea of early warning signals, since it is often useful to be forewarned of an impending catastrophe. The above list of nine shared tipping-point attributes references an “increase in variance” and an “increase in autocorrelation” prior to the tipping point, both of which facts are commonly used to predict a tipping point before it happens, given that the existence of a critical threshold is known [Held and Kleinen, 2004; Livina and Lenton, 2007; Scheffer et al., 2009; Thompson and Sieber, 2011]. Thus, it is possible to use measurements of variance and autocorrelation as tipping point indicators which may provide early warning signals of a future event. This is the basis of all the techniques used throughout this thesis, although we also use measurements of the power spectrum properties which are closely related to autocorrelation, and detrended fluctuation analysis (DFA), both of which are described in section ???. In order to see the change in these statistics over time we use a tipping point indicator [Held and Kleinen, 2004]: the ACF, variance, or other relevant statistic is calculated in overlapping segments of the time series. In chapter ??? three tipping point indicators (ACF, DFA, and the novel PS indicator) are applied to various time series.

A key concept involved in these methods is that of *resilience* or *recovery rates* [Scheffer et al., 2001; Veraart et al., 2012], that is, how quickly the system returns to the stable state after a small perturbation. The theory states that the recovery rate will slow down as the system approaches a bifurcation point [Scheffer et al., 2009], known as critical slowing down. This is equivalent to an increase in long-term correlations or *memory*.

As an example we consider the pitchfork bifurcation illustrated in figure 1.6. This system is trapped within a single well of attraction (at $z = 0$) for $t \leq 600$, at $t = 600$ a bifurcation occurs creating a double-well potential system for $t > 600$. For $t \ll 600$ the sides of the potential well are steep and the system quickly returns to the equilibrium $z = 0$ following a small perturbation due to noise. For $t = 600$ (panel **b**), however, this process takes much longer because the base of the well is flat. During the transition from panel **a** to panel **b** the system will take longer and longer to return to the $z = 0$ equilibrium following a small perturbation, as the sides of the potential well become less steep. This

increasing *return time*, or decreasing *resilience* is known as critical slowing down. Because of the generalness of the pitchfork bifurcation [Kuehn, 2011], the hypothesis is that critical slowing down will be observed in a wide variety of dynamical systems approaching a bifurcation [Scheffer et al., 2009]. This is tested by Van Nes and Scheffer [2007] who measure critical slowing down in a number of different ecological models and find the effect to occur universally.

The speed at which a system returns to equilibrium can be measured by the autocorrelation, since a non-retuning system will behave as a random walk, whereas a returning system is usually governed only by noise [Scheffer et al., 2009]. It is this observation which allows us to successfully apply autocorrelation-based tipping point indicators to several different dynamical systems in chapters ?? and ??, and to use these indicators to detect tipping points in time series of geophysical variables in chapter ??.

1.4 Outline of thesis

In chapter ?? we study different tipping point indicators which are applied to one-dimensional time series exhibiting tipping points. The indicators include the lag-1 autocorrelation function [Held and Kleinen, 2004; Dakos et al., 2008], the use of which has been mentioned in this introduction, the detrended fluctuations analysis (DFA) coefficient [Kantelhardt et al., 2001], which has been used previously in tipping point research [Livina and Lenton, 2007], and the power spectrum scaling exponent, the use of which in tipping point analysis is a novel outcome of this research [Prettyman et al., 2018].

In chapter ?? the tipping point indicators used in chapter ?? are adapted for use in multi-dimensional systems, both by evaluating the indicators simultaneously at many points over a field to provide a visual description of the indicator's behaviour, and also by reducing the dimension of a system using empirical orthogonal functions (EOFs) [Held and Kleinen, 2004]. The specific applicability of the EOF method to tipping point analysis is investigated analytically using a general dynamical system containing a tipping point, and experiments are also made with novel variations of the EOF method in a tipping points context. The methods presented are also compared to an existing method for producing early warning signals of tipping points in two-dimensional systems [Williamson and Lenton, 2015].

In chapter ?? the methods introduced throughout the thesis are applied to measurement data from a geophysical system, that is, to measurements of sea-level pressure at points close to the landfall locations of tropical cyclones. A stochastic model of an approaching tropical cyclone is also presented, modified from a simple existing model [Holland, 1980] and parametrised using the results from our tipping point analysis.

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